

Project Title: Travelog

Team 4

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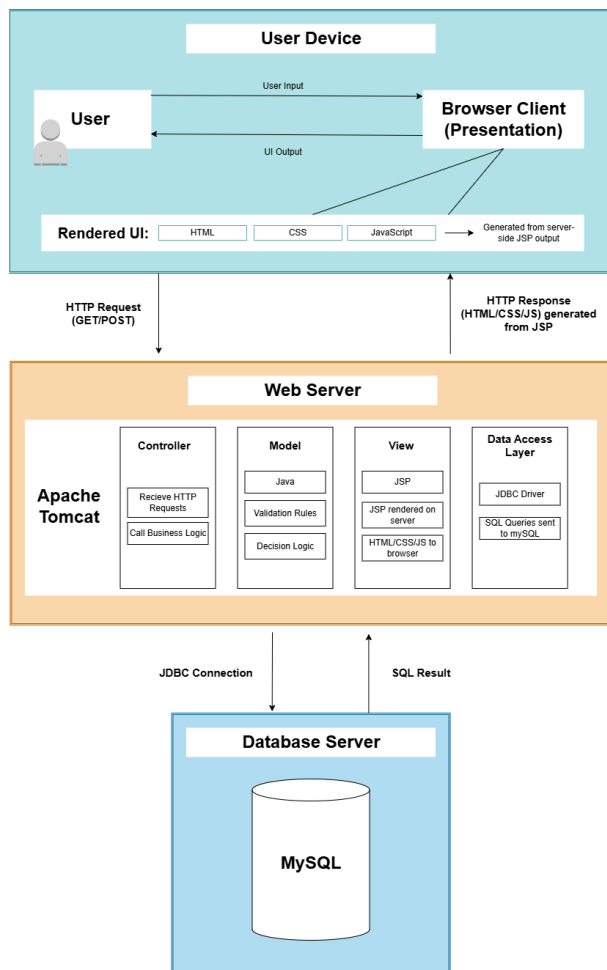
Project Description

Travelog is a travel management web application that allows users to organize, plan, book, and cancel trips all in one place.

Users can get started by creating an account and then adding a trip to their dashboard. With a new trip created, they can select a travel destination and view transportation and lodging options based on their travel date and budget preferences. They can then quickly purchase these key components of any trip immediately after they are done selecting and reviewing options. Furthermore, with this platform users can log multiple trips, cancel when plans change, and review all their bookings. Simply put, Travelog seamlessly integrates the hardest parts of travel planning into one user-friendly platform.

This application will also allow hotels and transportation companies to create corporate accounts and post unseen offers, availability, and discounts to Travelog users. These listings will include a wide range of information including cost and other details, such as hotel rating. Furthermore, administrators (i.e., support staff) will have robust and secure access control to help users with concerns, issues, and updates without privacy concerns.

System Environment



Hardware and Software: MacOS Laptop, Apache Tomcat 9.0.115, MySQL Workbench 8.0.46

RDBMS: MySQL Community Server 8.4.0

Application Languages: JSP, Java, SQL, XML, HTML, CSS

The diagram to the left was created with draw.io and can also be accessed with this [link](#).

Functional Requirements

For this application, we introduce three user views: Traveler, Company, and Administrator.

Traveler View:

Critically, on Travelog, a traveler is able to register, login, logout, and delete their account.

Register Account

Description:

A traveler can create a new account on Travelog. The system will collect registration information, validate it, and store a new record in the Users table on the database. Duplicate accounts will not be allowed via unique constraints on the username and the email account. For security purposes, the password will also be encrypted before being inputted into the database.

Input:

The registration form will have the following fields:

Username (String, Required, Unique)

Email Address (String, Required, Unique, Valid Email Format)

Password (String, Required, Encrypted Before Storage)

Process:

1. The traveler accesses the registration page and will enter the information into the form fields.
2. The system will validate that all required fields are complete
3. The system verifies that username and email address does not already exist in the Users table, the email format is valid, and that the password meets required security measures (e.g., length and alphanumeric constraints).
4. If validation passes, then the system will encrypt the password, and insert a new record into the users table containing a username, email, and encrypted password. A confirmation message will also be displayed to the user and they will be redirected to the homepage of the application.
5. If the validation fails, then the system will cancel the registration process and show the corresponding error message.

Account Sign In

Description:

A traveler can login to an existing account on Travelog. The system will authenticate the user by comparing the collected login information with the records in the Users table on the database. If the authentication is successful, then the user will be able to access the homepage via a secure session.

Input:

The login form will have the following fields:

Username (String, Required)

Password (String, Required)

Process:

1. The traveler will enter their username and password into the login fields.
2. The system will validate that all required fields are complete
3. The system will pull the matching record from the User table in the database.
4. If the corresponding record is found, the system will redirect the user to the homepage via a secure session.
5. If the corresponding record is not found, the system will display the error message and the login is cancelled.

Account Logout

Description:

A traveller can logout of their account and their secure session will be ended by the system.

Input:

Logout (Button)

Process:

1. The traveler will click the logout button.

2. The system will terminate the current session.
3. The traveler is redirected to the login page.

Delete Account

Description:

A traveller can delete their account and remove all their data and information from Travelog.

Input:

Password (String, Required)

Process:

1. Traveller will click the delete account button
2. The system will give a confirmation message where the user needs to input their password
3. The system will compare the password with records in the User table
4. If the corresponding match is found, the system will delete the record from the User table, and perform a cascading delete to remove any associated records.
5. If the match is not found (i.e., the password is incorrect), then the system will cancel deletion and show an error message.

On Travelog, a traveler is also able to view and create trips, and pick corresponding destinations and dates for each one.

View trips

Description:

A traveller can view all the trips they have created thus far on a personal dashboard. They can see their planned, booked and cancelled trips all in one place.

Input:

Secure User Session (No Manual Input)

Process:

1. The system will do an SQL query for all the trips under the current user's id in the User_Trips table.
2. If the query is empty, a message stating no trips created will be displayed.
3. If the query has results, the trips will be displayed on the dashboard for the user in descending order by status first (i.e., planned, booked, and cancelled), then by date.

Create trip

Description:

A traveller can create a new trip that will later contain all the trip details (i.e., destination, dates, accommodations, etc.)

Input:

Trip Name (String, Required)

Process:

1. Traveler will click create trip
2. Traveler will enter the trip's name
3. The system will then enter the trip as a new record in the User_Trips table.
4. The traveler will be redirected into the trip's homepage.

Select Starting and Final destinations

Description:

A traveller can select their starting origin and their destination for the trip they created.

Input:

Trip ID (No Manual Input)

Starting Location (String, Required, Pick From List or Search)

Final Destination (String, Required, Pick From List or Search)

Process:

1. Travelers will search or pick destinations. For a search, the system will do an internal query to match the search.
2. The traveler will select and confirm the origin and destination.

3. The system updates the corresponding user_trips row in the database and the changes show on the trip's homepage.

Select preferred dates

Description:

A traveller can select their preferred dates for the trip they created.

Input:

Trip ID (No Manual Input)
Start Date (Date, Required)
End Date (Date, Required)

Process:

1. The traveler picks the start and end dates using a date picker
2. The system validates that the end date is not before the start date and is after the current date.
3. The system updates the corresponding user_trips row in the database and the changes show on the trip's homepage.

On Travelog, a traveler is also able to browse and filter transport options, confirm their choices, and update transport plans to account for changes.

Browse and filter transport options

Description:

A traveller can view the different transport options like bus, plane, or train that work with the trip's date range and destination. Results can also be filtered based on certain criteria.

Input:

Trip ID (No Manual Input)
Selected Filters: Transport Type, Cost, Time (Optional)

Process:

1. The traveller searches in the search bar or selects filters.
2. The system will do a query for transport options based on the travellers search, selected filters, destination, and dates. If the user has saved preferences, it will be retrieved from the user_preferences table and applied.
3. The system will display all corresponding and filtered results.

Select transportation

Description:

A traveller can select their transportation for both trips on the journey (to and from).

Input:

Trip ID (No Manual Input)
Selected Transport (Transport ID (automatic based on selection), Required)
Direction (Dropdown, Required, Return or Outbound)

Process:

1. The traveller selects the transportation option they like and clicks confirm.
2. The system fills out a record with all the information and adds it to the trip_transport table.
3. If a transportation choice is already in the table, it will overwrite the previous record. This works as an update option for travellers.

On Travelog, a traveler is also able to browse and filter hotel options, confirm their choices, and update hotel plans to account for changes.

Browse and filter hotel options

The same logic as "Browse and filter transport options" applies here. Refer to the explanation for that functional requirements and replace transportation with hotel. Filters may be modified to number of beds, windows and other variables.

Select hotel

Description:

A traveller can select their hotel for their trip.

Input:

Trip ID (No Manual Input)

Selected Hotel (Hotel ID (automatic based on selection), Required)

Process:

1. The traveler selects a hotel and enters the specific dates.
2. The system validates the dates are feasible.
3. The system adds a record into the trip_hotels with the required information.

On Travelog, a traveler is able to confirm their booking and move their trip from a planned to booked status. They can also then cancel their booking, and move their booked status to cancelled. They may possibly receive a refund as well.

Review and Confirm Trip Booking

Description:

The traveler can view their cost summary of the trip (both hotel and transportation) and then finalize the booking.

Input:

Trip ID (No Manual Input)

Payment Method (Form, Required)

Process:

1. The system queries the Trip ID to get the correct information from the trip_transport and trip_hotels tables.
2. The costs pulled from the transportation and hotel tables will be summarized and calculated by the system.
3. The total summary will be shown to the traveller who can either continue with the booking or cancel and return to the trip homepage.
4. If they continue, the traveller selects their payment method and fills it out, here security is critical.
5. A new record will be created in the Payments table which will not contain any sensitive information but rather metadata about the traveller's payment for the booking.
6. The trip status is then updated from planned to booked.

Cancel Trip and Receive Refund

Description:

The traveller can cancel portions of the trip or the entire trip and receive a partial or full refund.

Input:

Trip ID (No Manual Input)

Cancellation Level (Dropdown, Required, Whole trip, hotel, or transportation)

Password (String, Required)

Process:

1. The traveler selects the level of which they are cancelling.
2. The system allows them to make a more detailed selection of what they are cancelling (i.e., one of the booked hotels or transportation).
3. The system calculates expected refunds, the traveller can then continue or cancel their cancellation.
4. The traveller enters their password to confirm their decision if they do cancel.
5. The system updates the status accordingly in all corresponding records (trips, trip_hotels, trip_transportations, payments).

Company View:

Critically, on Travelog companies can register, login, logout, and delete their account.

Register Account, Account Sign In, Account Signout, Delete Account

Fundamentally, there is no difference between this and the previously listed requirements for Travellers. Thus, to understand the register account, login, logout, and delete functionalities for companies refer to the traveler view above. One major change is that for an account created under a company, it will be manually audited by the Travelog personnel.

Furthermore, on Travelog, companies can post unseen offers, availability, and discounts for Travelog travelers. They can also delete any old or outdated offers.

Post Offer

Description:

Companies can post or update existing posts under their brand for Travelog users. These updates and offers could include things like discounts and new availability.

Input:

Destination (String, Required)
Name (String, Required)
Cost (String, Required)
Availabilities (String, Required)
Discounts (String, Required)

Process:

1. Company clicks on create or manage postings on the homepage.
2. Company enters modified information or new information for a posting.
3. System verifies all information is valid per constraints and expectations.
4. New postings will be added to the hotels or transportation tables by the system, updates will update existing records.
5. Confirmation is shown to the company once the posting is successfully added.

Delete Post

Description:

Companies can delete existing posts from Travelog so that users can no longer see it.

Input:

Posting ID (No Manual Input, Required)
Password (String, Required)

Process:

1. Company clicks on an existing post to delete.
2. Company clicks on delete confirmation.
3. The system prompts the company to enter a password. Validation by system occurs, failure cancels the deletion process.
4. If the company is successful in confirming deletion, the posting is removed from Travelog users except those who have already booked the accommodations or transportation.
5. The system archives records using a status column rather than a delete to avoid voiding existing users who have previously booked.

Administrator View:

Admins will have existing accounts, which they will simply use to login. Login functionality is equivalent to any regular user's experience logging in. Thus, the only important functionality these users have is confirming company accounts and handling user queries.

Confirm or Reject Company Accounts

Description:

Administrators can review to approve or deny a company account on the platform.

Input:

Selected Company (Click on Company)
Approval Choice (Accept or Deny, Required)

Process:

1. Administrator sees list of accounts awaiting approval and selects company for review.
2. The administrator chooses to deny or accept.
3. The status of the company account is changed by the system accordingly, and an email notification to the company is sent out regarding the decision.

4. The list of accounts is now reduced by one through system update of the list.

Handle User Queries

Description:

Administrators can deal with user accounts and issues sent through queries on the application.

Input:

Selected query (Required)

Administrative Action (Required)

Process:

1. Administrator selects a query in their inbox
2. The system retrieves the query from the Query table
3. The Administrator makes an action based on the query
4. The system records any changes made including updates to the database.

Non-functional Issues

Graphical User Interface (GUI)

Travelog is designed as a simple, user-friendly website to make it easy for travelers to plan and book trips. The website is designed with a simple interface that guides users through account creation, trip planning, and viewing trips, and it also allows users to cancel trips if they wish. We will be using JSP and Java to build the web application and HTML and CSS for the interface. Our backend will consist of MySQL and all this will be hosted on a Tomcat server.

Screens

Landing page

- Consists of logo/Title
- Button that allows the user to start creating trip/ will prompt to login or create an account
- Grid of destination pictures in the center
- Login button top right

Login/Register

- Username/Password fields
- Login button
- Register button
- Feedback message if Incorrect username or Password

Dashboard

- Displays user trips
- Create new trip button

Trip builder

- Dropdown menu/search bar for searching destinations
- Menu for inputting user preferences
- Button to start search
 - After button pressed, listings are displayed
 - Listings will have a dropdown menu for saving and booking

Flight Selection

- Based on destination, it will list the flight
- Show price and time
- Filter button (time, layover, cost)

Hotel Selection

- Name
- Rating
- Price
- Filter button (rating, cost)

Review/Confirm Page

- Order summary / Flight/ Hotel
- Total cost of trip
- Confirmation button

Access control

Users who access the website can view destinations, but will be prompted to create an account before viewing hotel/transportation information and saving any travel plans. In order to book or cancel a trip, a user will need to confirm their decision using their password. Planning trips does not require password confirmation.

Once an account is created, hotel and transportation services can update availability through their accounts. They are able to modify the company name and availability for services. These service accounts can only view user information from bookings. User draft trips will not be available for services to view.

- Users (Not Logged in)
 - Can browse destinations
 - Cannot book trips nor will any data be saved
- Users Logged in
 - Can create a trip plan, view trip, and cancel
 - Can book and data will be stored in the database
 - Cannot see any other user data, only their trip
- Companies
 - Can create and manage listings, but only their own
 - Can update listing
 - Cannot access any other data but theirs
 - Can remove listing
- Admins
 - Have full access to manage all

Security

Account credentials

- User authentication is primarily through a username/password system. Passwords will not be stored in plain text, but rather with a hashing system to protect personal information.
- Requires Users to input a unique username and email so there are no duplicates
- Passwords will require a level of complexity in order to stop any malicious behavior on the website
- If any cancellation happens, it will require confirmation in order to cancel the trip

Authentication

- User must log in to view, book, or cancel trips